

IT Fundamentals+ Certification

300 Interview Questions & Answers

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How to use this guide: Each question is followed by a comprehensive answer (▢). Important exam tips and notes are highlighted (▢). All 10 sections together cover the full CompTIA ITF+ FC0-U61 syllabus.

SECTION 01

IT Concepts & Terminology

Q1 – Q30

Q1 What is Information Technology (IT)?

Answer: IT refers to the use of computers, software, networks, and other digital infrastructure to store, process, transmit, and retrieve data. It encompasses all forms of technology used to create, exchange, and use information in business and personal contexts.

Q2 What is hardware?

Answer: Hardware refers to all the physical components of a computer system, including the CPU, RAM, hard drives, keyboard, mouse, monitor, motherboard, and peripheral devices. These are tangible components you can physically touch and see.

Q3 What is software?

Answer: Software is a collection of programs, instructions, and data that tells hardware what to do. It is intangible and includes operating systems (like Windows, Linux), applications (like MS Word, Chrome), and utilities.

Q4 What is the difference between system software and application software?

Answer: System software manages hardware and provides a platform for applications. Examples: OS, BIOS, device drivers. Application software is designed for end-users to perform specific tasks. Examples: MS Office, Photoshop, web browsers.

Q5 What is an operating system (OS)?

Answer: An OS is system software that manages hardware resources, provides services for applications, and acts as an interface between users and hardware. Examples: Windows 11, macOS, Linux (Ubuntu), Android, iOS.

Q6 What is the CPU and what does it do?

Answer: CPU (Central Processing Unit) is the primary component that executes instructions of computer programs. It performs arithmetic, logic, control, and I/O operations. Speed is measured in GHz (gigahertz). Examples: Intel Core i7, AMD Ryzen 5.

Q7 What is RAM and why is it important?

Answer: RAM (Random Access Memory) is volatile short-term memory that temporarily stores data currently being used by the CPU. More RAM allows more programs to run simultaneously. Common sizes: 8 GB, 16 GB, 32 GB. It is lost when power is off.

Q8 What is ROM?

Answer: ROM (Read-Only Memory) is non-volatile memory that permanently stores firmware and BIOS instructions. Unlike RAM, data in ROM is not lost when power is off. It cannot be easily modified by users.

Q9 What is a hard disk drive (HDD)?

Answer: HDD is a traditional mechanical storage device that uses magnetic spinning platters to store data permanently. HDDs have high capacity but are slower compared to SSDs. They are used for long-term storage of files, OS, and programs.

Q10 What is an SSD and how is it different from HDD?

Answer: SSD (Solid-State Drive) uses flash memory chips with no moving parts, making it significantly faster, quieter, and more durable than HDD. SSDs are more expensive per GB but offer much better performance. Modern laptops prefer SSDs.

Q11 What is the motherboard?

Answer: The motherboard is the main printed circuit board (PCB) that connects and allows communication between all components: CPU, RAM, storage, GPU, and other peripherals. It contains the chipset, expansion slots, and BIOS chip.

Q12 What is BIOS/UEFI?

Answer: BIOS (Basic Input/Output System) or UEFI (Unified Extensible Firmware Interface) is firmware stored on a chip on the motherboard. It initializes hardware during boot, runs POST (Power-On Self-Test), and loads the operating system.

Q13 What is a GPU?

Answer: GPU (Graphics Processing Unit) is a specialized processor designed to accelerate graphics rendering. It handles display output and is critical for gaming, video editing, and AI workloads. Examples: NVIDIA RTX 4070, AMD Radeon RX 7800.

Q14 What is binary number system?

Answer: Binary is a base-2 number system using only 0 and 1. Computers use binary because digital circuits have two states: on (1) and off (0). All data in computers is ultimately stored and processed as binary.

Q15 What is a bit and a byte?

Answer: A bit is the smallest unit of digital data, representing either 0 or 1. A byte consists of 8 bits. Larger units: 1 KB = 1024 bytes, 1 MB = 1024 KB, 1 GB = 1024 MB, 1 TB = 1024 GB.

Q16 What is the difference between 32-bit and 64-bit systems?

Answer: 32-bit systems can address maximum 4 GB of RAM, while 64-bit systems can address terabytes of RAM. 64-bit systems are faster and can run both 32-bit and 64-bit applications. Most modern systems are 64-bit.

Q17 What is an input device? Give examples.

Answer: Input devices allow users to enter data into the computer. Examples: keyboard, mouse, touchscreen, scanner, microphone, webcam, stylus pen, gamepad, barcode reader.

Q18 What is an output device? Give examples.

Answer: Output devices display or produce results from the computer. Examples: monitor, printer, speakers, projector, headphones, plotter.

Q19 What is a peripheral device?

Answer: Peripheral devices are external hardware components connected to the computer to expand its functionality. They can be input (keyboard), output (monitor), or storage (external HDD) devices.

Q20 What is virtual memory?

Answer: Virtual memory is a memory management technique that uses a portion of the hard drive as if it were RAM when physical RAM is full. It uses a paging file (pagefile.sys in Windows). It is slower than RAM but allows running larger applications.

Q21 What is the difference between volatile and non-volatile memory?

Answer: Volatile memory loses its data when power is removed (e.g., RAM, cache). Non-volatile memory retains data even without power (e.g., ROM, HDD, SSD, flash drives).

Q22 What is cache memory?

Answer: Cache memory is ultra-fast memory located close to or inside the CPU that stores frequently accessed data. L1 cache is fastest and smallest; L3 is larger but slower. Cache significantly reduces CPU waiting time.

Q23 What is firmware?

Answer: Firmware is permanent software programmed into read-only memory. It provides low-level control of hardware and typically does not change. Examples: BIOS, printer firmware, router firmware.

Q24 What is a file system?

Answer: A file system organizes and manages how data is stored and retrieved on storage devices. Examples: NTFS (Windows), FAT32, exFAT, ext4 (Linux), APFS (macOS). It manages file names, folders, permissions, and metadata.

Q25 What is the difference between FAT32 and NTFS?

Answer: FAT32 supports maximum file size of 4 GB and is compatible with most devices. NTFS supports very large files, has better security (permissions, encryption), journaling, and is the default file system for Windows.

Q26 What is an API?

Answer: API (Application Programming Interface) is a set of rules and protocols that allows different software applications to communicate with each other. APIs define how requests are made and how responses are given. Examples: REST API, Windows API.

Q27 What is open source software?

Answer: Open source software has publicly available source code that anyone can view, modify, and distribute. Examples: Linux, Firefox, LibreOffice, VLC. It contrasts with proprietary software where source code is not available.

Q28 What is a driver?

Answer: A device driver is software that allows the OS to communicate with hardware devices. Each hardware component needs a compatible driver. Without drivers, devices cannot function properly. Examples: printer driver, graphics driver.

Q29 What is multitasking in computing?

Answer: Multitasking is the ability of an OS to run multiple processes or applications simultaneously. The CPU rapidly switches between tasks giving an impression of parallel execution. Modern multi-core CPUs can truly run tasks in parallel.

Q30 What is the difference between a process and a thread?

Answer: A process is an independent program in execution with its own memory space. A thread is a smaller unit of execution within a process. Multiple threads within a process share the same memory, enabling faster communication and parallel execution.

SECTION 02

Networking Concepts

Q31 – Q60

Q31 What is a computer network?

Answer: A computer network is a group of two or more interconnected devices that share resources and communicate. Networks are classified by size: LAN, MAN, WAN, and by topology: star, bus, ring, mesh.

Q32 What is the difference between LAN, MAN, and WAN?

Answer: LAN (Local Area Network) covers a small area like a building or office. MAN (Metropolitan Area Network) covers a city. WAN (Wide Area Network) covers large geographic areas; the Internet is the largest WAN.

Q33 What is an IP address?

Answer: An IP address is a unique numerical label assigned to each device on a network for identification and location. IPv4 uses 32-bit addresses (e.g., 192.168.1.1). IPv6 uses 128-bit addresses for greater availability.

Q34 What is the difference between IPv4 and IPv6?

Answer: IPv4 uses 32-bit addresses allowing ~4.3 billion unique addresses. IPv6 uses 128-bit addresses allowing virtually unlimited addresses, written in hexadecimal. IPv6 was created to solve IPv4 address exhaustion.

Q35 What is a MAC address?

Answer: MAC (Media Access Control) address is a unique hardware identifier assigned to a network interface card (NIC) by the manufacturer. It is a 48-bit address written as 6 hexadecimal pairs (e.g., 00:1A:2B:3C:4D:5E). It operates at Layer 2.

Q36 What is DHCP?

Answer: DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network configuration to devices. A DHCP server leases IP addresses dynamically, eliminating manual configuration. Default ports: 67 (server), 68 (client).

■ *Key in your FortiGate/network admin role – DHCP scope management is daily work.*

Q37 What is DNS?

Answer: DNS (Domain Name System) translates human-readable domain names (like `www.google.com`) into IP addresses that computers use to communicate. It acts as a phone book of the internet. Default port: 53.

Q38 What is a router?

Answer: A router is a network device that forwards data packets between different networks. It determines the best path for data using routing tables and protocols. Routers operate at Layer 3 (Network layer) of the OSI model.

Q39 What is a switch?

Answer: A switch connects multiple devices within the same LAN. It operates at Layer 2, uses MAC addresses to forward frames to the correct device, and creates separate collision domains for each port — more efficient than a hub.

Q40 What is a hub?

Answer: A hub is a basic networking device that broadcasts data to all ports regardless of destination, creating collisions and reducing efficiency. Hubs are mostly obsolete, replaced by switches in all modern networks.

Q41 What is the OSI model?

Answer: OSI (Open Systems Interconnection) model is a conceptual framework with 7 layers that standardizes how different systems communicate. It helps troubleshoot and design networks systematically.

Q42 What are the 7 layers of the OSI model?

Answer: Layer 1: Physical (cables, hubs), Layer 2: Data Link (switches, MAC), Layer 3: Network (routers, IP), Layer 4: Transport (TCP/UDP), Layer 5: Session, Layer 6: Presentation (encryption), Layer 7: Application (HTTP, FTP, DNS).

■ Mnemonic: 'Please Do Not Throw Sausage Pizza Away'

Q43 What is TCP/IP model?

Answer: TCP/IP model has 4 layers: Network Access (combines OSI L1+L2), Internet (OSI L3), Transport (OSI L4), and Application (OSI L5-L7). It is the practical model used in real-world networking and the Internet.

Q44 What is the difference between TCP and UDP?

Answer: TCP is connection-oriented, reliable, ensures ordered delivery with acknowledgments. Used for web browsing, email, file transfer. UDP is connectionless, faster, no guaranteed delivery. Used for video streaming, VoIP, gaming.

Q45 What is a firewall?

Answer: A firewall monitors and controls incoming and outgoing network traffic based on predefined security rules. It creates a barrier between trusted internal networks and untrusted external networks. Can be hardware or software-based.

■ *Directly relevant to your FortiGate experience!*

Q46 What is NAT?

Answer: NAT (Network Address Translation) maps private IP addresses to a public IP address when sending traffic to the internet. It conserves public IPs and provides a layer of security by hiding internal network structure.

Q47 What is a subnet mask?

Answer: A subnet mask defines the network and host portions of an IP address. Example: 255.255.255.0 (/24) means the first 3 octets identify the network and the last octet identifies hosts — allowing 254 usable host addresses.

Q48 What is the default gateway?

Answer: The default gateway is the IP address of the router that routes traffic from the local network to other networks or the internet. It is the exit point for packets destined outside the local subnet.

Q49 What is Wi-Fi?

Answer: Wi-Fi is a wireless networking technology using radio waves to provide internet and network connectivity. It follows IEEE 802.11 standards. Common standards: 802.11n/ac/ax (Wi-Fi 6). Operates on 2.4 GHz and 5 GHz bands.

Q50 What is the difference between 2.4 GHz and 5 GHz Wi-Fi?

Answer: 2.4 GHz has longer range but slower speed and more interference from other devices. 5 GHz has shorter range but faster speed and less congestion. Wi-Fi 6 (802.11ax) improves performance on both bands simultaneously.

Q51 What is Ethernet?

Answer: Ethernet is the most common wired networking standard (IEEE 802.3) using cables (Cat5e, Cat6, Cat6a) to connect devices. Speeds range from 10 Mbps to 100 Gbps. It is reliable, secure, and faster than wireless for stable connections.

Q52 What is a VPN?

Answer: VPN (Virtual Private Network) creates an encrypted tunnel over the internet to securely connect users to a remote network. It masks the user's IP, ensures privacy, and is used by remote workers to access corporate networks securely.

Q53 What is a port number?

Answer: Port numbers identify specific applications or services on a network device. Range: 0-65535. Well-known ports: HTTP=80, HTTPS=443, FTP=21, SSH=22, DNS=53, SMTP=25, RDP=3389, DHCP=67/68.

Q54 What is HTTP vs HTTPS?

Answer: HTTP transfers web pages without encryption on port 80. HTTPS uses SSL/TLS encryption on port 443, ensuring data confidentiality and integrity. HTTPS is required for all secure websites handling passwords or payments.

Q55 What is bandwidth?

Answer: Bandwidth is the maximum rate of data transfer across a network path, measured in bits per second (bps, Mbps, Gbps). Higher bandwidth allows more data to be transmitted simultaneously. It is different from latency.

Q56 What is latency?

Answer: Latency is the time delay between sending a request and receiving a response, measured in milliseconds (ms). High latency causes lag in real-time applications like gaming and VoIP. Affected by distance, congestion, and routing hops.

Q57 What is an ISP?

Answer: ISP (Internet Service Provider) is a company that provides internet access. Examples: Jio, Airtel, BSNL (India), Comcast (USA). ISPs connect users to the broader internet infrastructure and typically provide static or dynamic IP addresses.

Q58 What is a NIC?

Answer: NIC (Network Interface Card) is hardware that enables a computer to connect to a network. It can be wired (Ethernet port) or wireless (Wi-Fi adapter). Each NIC has a unique MAC address assigned by the manufacturer.

Q59 What is ping and what does it test?

Answer: Ping sends ICMP echo request packets to test connectivity between two devices. It measures round-trip time (RTT) in milliseconds. High ping = high latency. Example command: 'ping google.com' or 'ping 8.8.8.8'

Q60 What is the difference between intranet and internet?

Answer: The Internet is a global public network accessible to everyone worldwide. An intranet is a private network within an organization using internet technologies, accessible only to authorized users. An extranet extends intranet access to select external partners.

SECTION 03

Security Concepts

Q61 – Q90

Q61 What is cybersecurity?

Answer: Cybersecurity is the practice of protecting computers, networks, programs, and data from unauthorized access, damage, or attack. It encompasses network security, application security, information security, and operational security.

Q62 What is the CIA Triad?

Answer: CIA Triad is the foundation of information security: Confidentiality (only authorized users can access data), Integrity (data is accurate and untampered), and Availability (data and systems are accessible when needed by authorized users).

- *This is the most fundamental security concept – always mention it first in security interviews.*

Q63 What is authentication?

Answer: Authentication verifies the identity of a user, device, or system. Methods: something you know (password/PIN), something you have (smart card/token), something you are (biometrics), and somewhere you are (location-based).

Q64 What is Multi-Factor Authentication (MFA)?

Answer: MFA requires two or more verification factors to gain access. Example: password + OTP on phone. It significantly reduces the risk of unauthorized access even if one factor (like a password) is compromised.

Q65 What is encryption?

Answer: Encryption converts readable data (plaintext) into unreadable format (ciphertext) using an algorithm and key. Only parties with the correct decryption key can read the data. Key algorithms: AES (symmetric), RSA (asymmetric).

Q66 What is a digital certificate?

Answer: A digital certificate is an electronic document that verifies the identity of a website, person, or device. Issued by a Certificate Authority (CA), it contains the public key and is used in HTTPS to establish secure connections.

Q67 What is malware?

Answer: Malware (malicious software) is any software designed to harm, exploit, or attack systems. Types include: viruses, worms, trojans, ransomware, spyware, adware, rootkits, and keyloggers.

Q68 What is a virus?

Answer: A computer virus attaches to legitimate files and spreads when infected files are shared. It needs a host file to propagate and can damage or delete data, corrupt the OS, or display unwanted messages.

Q69 What is a worm?

Answer: A worm is self-replicating malware that spreads across networks without needing a host file or user action. It consumes bandwidth and can carry destructive payloads. Famous examples: WannaCry, ILOVEYOU.

Q70 What is a Trojan horse?

Answer: A Trojan is malware disguised as legitimate software. It does not self-replicate but tricks users into installing it, then creates backdoors, steals data, or causes damage. Named after the ancient Greek myth.

Q71 What is ransomware?

Answer: Ransomware encrypts the victim's files and demands payment (ransom) to restore access. Delivered via phishing emails or malicious downloads. Famous examples: WannaCry, REvil. Prevention: regular backups, patch management, email filtering.

- *Ransomware attacks increased 150%+ in recent years — hot interview topic in cybersecurity.*

Q72 What is phishing?

Answer: Phishing sends fraudulent emails or messages pretending to be legitimate organizations to trick users into revealing passwords, credit card numbers, or other sensitive data. Most common entry point for corporate breaches.

Q73 What is spear phishing?

Answer: Spear phishing is a targeted phishing attack directed at specific individuals or organizations, using personalized information to appear credible. Far more dangerous than generic mass phishing attacks.

Q74 What is social engineering?

Answer: Social engineering is the psychological manipulation of people into performing actions or divulging confidential information. It exploits human psychology, not technical vulnerabilities. Examples: phishing, pretexting, baiting, tailgating, vishing.

Q75 What is a firewall and its types?

Answer: A firewall controls network traffic based on security rules. Types: Packet filtering (examines headers), Stateful inspection (tracks connections), Application-layer/Proxy (inspects content), NGFW (includes IPS, deep packet inspection, and application awareness).

Q76 What is antivirus software?

Answer: Antivirus detects, prevents, and removes malware. It uses signature-based detection (known malware), heuristic analysis (suspicious behavior), and real-time protection. Examples: Windows Defender, Norton, Kaspersky, CrowdStrike.

Q77 What is a DDoS attack?

Answer: DDoS (Distributed Denial of Service) floods a target with massive traffic from multiple sources (botnet) to make it unavailable to legitimate users. Mitigation includes CDNs, rate limiting, and DDoS protection services like Cloudflare.

Q78 What is a man-in-the-middle attack?

Answer: MITM occurs when an attacker secretly intercepts and possibly alters communication between two parties who believe they are communicating directly. Prevention: encryption (HTTPS/TLS), VPN, certificate pinning, proper certificate verification.

Q79 What is SQL injection?

Answer: SQL injection inserts malicious SQL code into input fields to manipulate the database — exposing, modifying, or deleting data. Prevention: parameterized queries (prepared statements), input validation, and stored procedures.

Q80 What is a brute force attack?

Answer: A brute force attack systematically tries all possible password combinations until the correct one is found. Prevention: account lockout policies, strong password requirements, MFA, and CAPTCHA challenges.

Q81 What is a password policy?

Answer: A password policy defines rules for creating and managing passwords. Best practices: minimum 12 characters, mix of uppercase, lowercase, numbers, symbols; avoid dictionary words; regular expiration; no reuse of recent passwords.

Q82 What is least privilege principle?

Answer: The Principle of Least Privilege (PoLP) states that users, processes, and systems should have only the minimum permissions necessary to perform their tasks. This limits the blast radius of accidents, errors, or attacks.

Q83 What is a vulnerability?

Answer: A vulnerability is a weakness or flaw in a system, software, or process that can be exploited. Vulnerabilities are cataloged in the CVE (Common Vulnerabilities and Exposures) database and scored using CVSS (Common Vulnerability Scoring System).

Q84 What is patch management?

Answer: Patch management is the systematic process of identifying, testing, and deploying software updates (patches) to fix security vulnerabilities and bugs. It is one of the most effective defenses against known exploits.

Q85 What is two-factor authentication (2FA)?

Answer: 2FA requires two forms of identification: typically a password (something you know) plus a one-time code sent to a phone or generated by an authenticator app (something you have). Reduces account compromise risk drastically.

Q86 What is a DMZ in networking?

Answer: DMZ (Demilitarized Zone) is a separate network segment between the internal trusted network and the untrusted internet. Publicly accessible servers (web, email, DNS) are placed in the DMZ, protecting the internal network from direct exposure.

Q87 What is data backup and the 3-2-1 rule?

Answer: Data backup copies data to another location to protect against loss from hardware failure, ransomware, or human error. The 3-2-1 rule: keep 3 copies of data, on 2 different media types, with 1 copy stored offsite.

Q88 What is the difference between IDS and IPS?

Answer: IDS (Intrusion Detection System) monitors network traffic and alerts on suspicious activity but does NOT block it — passive monitoring. IPS (Intrusion Prevention System) monitors AND actively blocks malicious traffic in real-time.

Q89 What is HTTPS and SSL/TLS?

Answer: HTTPS uses SSL/TLS (Transport Layer Security) protocols to encrypt data between a web browser and server. TLS is the modern, more secure successor to SSL. The padlock icon in browsers indicates an active HTTPS connection.

Q90 What is physical security in IT?

Answer: Physical security protects hardware and data from physical threats. Measures: access control systems (keycards, biometrics), security cameras, server room locks, cable locks for laptops, visitor logs, and environmental controls (fire, flood).

SECTION 04

Software & Programming
Fundamentals

Q91 – Q120

Q91 What is a programming language?

Answer: A programming language is a formal set of instructions to create software. It has syntax (grammar rules) and semantics (meaning). Types: compiled (C, C++), interpreted (Python, JavaScript), and scripted (Bash, PowerShell).

Q92 What is the difference between compiled and interpreted languages?

Answer: Compiled languages (C, C++, Java) are translated entirely to machine code before execution — faster performance. Interpreted languages (Python, Ruby) execute line-by-line at runtime — easier to debug but slightly slower.

Q93 What is an algorithm?

Answer: An algorithm is a step-by-step set of instructions to solve a problem. Algorithms must be finite, deterministic, and effective. They are analyzed for time complexity (Big O) and space complexity. Examples: binary search, quicksort.

Q94 What is a variable in programming?

Answer: A variable is a named storage location in memory holding a value that can change during execution. Variables have a data type, a name, and a value. Example: `int age = 25;` — declares an integer variable named 'age' with value 25.

Q95 What are the basic data types in programming?

Answer: Common data types: Integer (whole numbers), Float/Double (decimals), Boolean (true/false), Character (single char), String (text), Array (collection of same type), Object (complex structure combining data and methods).

Q96 What is a loop in programming?

Answer: A loop repeats a block of code until a condition is met. Types: for loop (fixed iterations), while loop (condition-based, checked before entry), do-while (executes at least once, condition checked after). Loops prevent code duplication.

Q97 What is a conditional statement?

Answer: Conditional statements execute code based on whether a condition is true or false. Types: if-else (binary choice), if-elif-else (multiple conditions), switch-case (multiple discrete values). Essential for program logic and decision-making.

Q98 What is a function in programming?

Answer: A function is a reusable, named block of code that performs a specific task. Functions accept input parameters, execute logic, and optionally return a value. They promote code reusability, modularity, testability, and readability.

Q99 What is object-oriented programming (OOP)?

Answer: OOP is a paradigm organizing code around 'objects' containing data (attributes) and behaviors (methods). Core principles (EIPA): Encapsulation, Inheritance, Polymorphism, and Abstraction. Examples: Java, Python, C++, C#.

Q100 What is a class and an object in OOP?

Answer: A class is a blueprint defining attributes and methods. An object is a specific instance of a class. Example: 'Car' is a class defining model, speed, accelerate(); 'myCar = new Car()' creates an object with its own state.

Q101 What is debugging?

Answer: Debugging identifies, analyzes, and fixes errors (bugs) in code. Methods: print statements, IDE debugger with breakpoints, code review, unit testing, and logging. IDEs like VS Code, IntelliJ, and PyCharm have powerful built-in debuggers.

Q102 What is a database?

Answer: A database is an organized collection of structured data stored electronically for easy access, management, and updating. Types: relational (MySQL, PostgreSQL, SQL Server) and non-relational/NoSQL (MongoDB, Redis, Cassandra).

Q103 What is SQL?

Answer: SQL (Structured Query Language) is the standard language for managing relational databases. Key commands: SELECT (retrieve data), INSERT (add rows), UPDATE (modify rows), DELETE (remove rows), CREATE TABLE, DROP TABLE.

Q104 What is the difference between SQL and NoSQL?

Answer: SQL databases use tables with fixed schema, support ACID transactions, and excel at complex queries on structured data. NoSQL uses flexible schemas (documents, key-value, graphs), scales horizontally, and suits unstructured or large-scale data.

Q105 What is version control?

Answer: Version control tracks changes to files over time, allowing developers to review history, revert to previous versions, and collaborate without overwriting each other's work. Git is the most popular distributed version control system.

Q106 What is Git?

Answer: Git is a distributed version control system tracking source code changes. Essential commands: git init, git clone, git add, git commit, git push, git pull, git branch, git merge. Platforms: GitHub, GitLab, Bitbucket.

Q107 What is a web browser?

Answer: A web browser retrieves and displays web pages from the internet. It interprets HTML, CSS, and JavaScript. Examples: Chrome, Firefox, Edge, Safari. Browsers use HTTP/HTTPS protocols to communicate with web servers.

Q108 What is HTML?

Answer: HTML (HyperText Markup Language) is the standard language for structuring web pages. It uses tags (html, head, body, h1-h6, p, a, img, div) to define content structure. HTML provides structure; CSS handles presentation.

Q109 What is CSS?

Answer: CSS (Cascading Style Sheets) controls how HTML elements are displayed — layout, colors, fonts, spacing, and responsive design. CSS separates content structure (HTML) from visual presentation, making maintenance easier.

Q110 What is JavaScript?

Answer: JavaScript is a scripting language that makes web pages interactive. It runs in the browser, manipulates HTML/CSS, handles events, validates forms, and fetches data from servers (AJAX/Fetch API). Also used server-side via Node.js.

Q111 What is an IDE?

Answer: IDE (Integrated Development Environment) provides tools for writing, testing, and debugging code: code editor with syntax highlighting, compiler/interpreter, debugger, and build automation. Examples: VS Code, IntelliJ IDEA, Eclipse, PyCharm.

Q112 What is front-end vs back-end development?

Answer: Front-end (client-side) is everything users see: HTML, CSS, JavaScript, React, Angular. Back-end (server-side) handles business logic and databases: Python, Node.js, Java, PHP. Full-stack developers work on both sides.

Q113 What is cloud computing?

Answer: Cloud computing delivers computing services (servers, storage, databases, software) over the internet on a pay-per-use basis. Service models: IaaS, PaaS, SaaS. Providers: AWS, Microsoft Azure, Google Cloud (GCP).

Q114 What is SaaS?

Answer: SaaS (Software as a Service) delivers software applications over the internet on subscription. Users access software via browser without installation. Examples: Google Workspace, Microsoft 365, Salesforce, Zoom, Dropbox.

Q115 What is IaaS?

Answer: IaaS (Infrastructure as a Service) provides virtualized computing resources: virtual machines, storage, networking. Users manage OS and applications; the provider manages physical hardware. Examples: AWS EC2, Azure VMs, Google Compute Engine.

Q116 What is PaaS?

Answer: PaaS (Platform as a Service) provides a platform for developers to build, test, and deploy applications without managing underlying infrastructure. Examples: Google App Engine, Heroku, Microsoft Azure App Service, AWS Elastic Beanstalk.

Q117 What is virtualization?

Answer: Virtualization creates virtual versions of computing resources using software (hypervisor), allowing multiple virtual machines to run on one physical server. Improves resource utilization, reduces hardware costs. Examples: VMware vSphere, VirtualBox, Hyper-V.

Q118 What is a hypervisor?

Answer: A hypervisor (VMM) creates and manages virtual machines. Type 1 (bare-metal) runs directly on hardware for best performance — VMware ESXi, Microsoft Hyper-V. Type 2 (hosted) runs on top of an OS — VirtualBox, VMware Workstation.

Q119 What is DevOps?

Answer: DevOps combines software development and IT operations to shorten development cycles and deliver high-quality software continuously. Key practices: CI/CD pipelines, infrastructure as code, automated testing, monitoring. Tools: Jenkins, Docker, Ansible, Git.

Q120 What is an operating system kernel?

Answer: The kernel is the core of the OS managing system resources: CPU scheduling, memory management, device I/O, and system calls. It operates in privileged mode (kernel space) with direct hardware access, separate from user-space applications.

SECTION 05

Database Fundamentals

Q121 – Q150

Q121 What is a relational database?

Answer: A relational database organizes data into tables (relations) with rows and columns, linked via primary and foreign keys. It uses SQL for manipulation. Examples: MySQL, Oracle, PostgreSQL, Microsoft SQL Server.

Q122 What is a primary key?

Answer: A primary key uniquely identifies each row in a table. It must be unique and NOT NULL. Every table should have a primary key for data integrity and efficient querying. Can be a single column or a composite of multiple columns.

Q123 What is a foreign key?

Answer: A foreign key is a column in one table that references the primary key of another table, establishing a relationship between them. It enforces referential integrity, preventing orphaned records in child tables.

Q124 What is normalization?

Answer: Normalization organizes a database to reduce data redundancy and improve integrity by dividing large tables into smaller, related ones. Normal forms: 1NF (atomic values), 2NF (no partial dependency), 3NF (no transitive dependency).

Q125 What is ACID in databases?

Answer: ACID properties ensure reliable transactions: Atomicity (all-or-nothing), Consistency (data remains valid after transaction), Isolation (concurrent transactions don't interfere), Durability (committed changes are permanent even after crash).

Q126 What is a SELECT statement?

Answer: SELECT retrieves data from tables. Syntax: SELECT columns FROM table WHERE condition ORDER BY column LIMIT n. SELECT * retrieves all columns. WHERE filters rows. ORDER BY sorts. JOIN combines multiple tables.

Q127 What is JOIN in SQL?

Answer: JOIN combines rows from multiple tables based on related columns. Types: INNER JOIN (only matching rows), LEFT JOIN (all from left + matches from right), RIGHT JOIN (all from right), FULL OUTER JOIN (all rows from both tables).

Q128 What is a database index?

Answer: A database index is a data structure that speeds up data retrieval by providing quick lookup paths, similar to a book index. Trade-off: faster SELECT queries but slightly slower INSERT/UPDATE/DELETE operations and extra storage.

Q129 What is a stored procedure?

Answer: A stored procedure is a precompiled set of SQL statements saved in the database, executed on demand. Benefits: reusability, enhanced security (users execute without direct table access), better performance, reduced network traffic.

Q130 What is a view in a database?

Answer: A view is a virtual table based on a SELECT query result. It does not store data physically. Views simplify complex queries, restrict access to sensitive columns, and present a consistent interface even when underlying tables change.

Q131 What is a NoSQL database?

Answer: NoSQL databases store data in non-tabular formats for flexible, scalable storage. Types: document (MongoDB), key-value (Redis), column-family (Cassandra), graph (Neo4j). Best for unstructured data, high throughput, and horizontal scaling.

Q132 What is MongoDB?

Answer: MongoDB is a document-oriented NoSQL database storing data as JSON-like BSON documents. It is schema-flexible, horizontally scalable, and popular for web and mobile applications needing dynamic data models. Uses collections instead of tables.

Q133 What is data redundancy?

Answer: Data redundancy is the storage of the same data in multiple places. While intentional redundancy (backups, RAID) improves reliability, uncontrolled redundancy wastes storage, creates inconsistency, and complicates updates. Normalization reduces it.

Q134 What is an ERD?

Answer: ERD (Entity-Relationship Diagram) visually represents database structure: entities (tables), attributes (columns), and relationships (connections). Cardinality notation shows one-to-one, one-to-many, and many-to-many relationships. Used in database design.

Q135 What is data integrity?

Answer: Data integrity ensures accuracy, consistency, and reliability of data. Types: Entity integrity (unique PKs), Referential integrity (valid FKs), Domain integrity (valid data types/ranges/constraints), and User-defined integrity (custom business rules).

Q136 What is a database transaction?

Answer: A transaction is a unit of work performed as a whole — all operations succeed or none do. Managed with COMMIT (save permanently) and ROLLBACK (undo changes). Essential for financial systems and any multi-step operations.

Q137 What is database backup?

Answer: Database backup copies data and structure to another location for recovery. Types: Full backup (entire database), Differential (changes since last full backup), Incremental (changes since last backup of any type — smallest and fastest).

Q138 What is a data warehouse?

Answer: A data warehouse is a large repository integrating data from multiple sources for analysis and reporting. Optimized for complex read queries (OLAP). Examples: Amazon Redshift, Google BigQuery, Snowflake. Distinct from OLTP operational databases.

Q139 What is OLTP vs OLAP?

Answer: OLTP (Online Transaction Processing) handles real-time transactions (INSERT/UPDATE/DELETE) with high concurrency — banking, e-commerce. OLAP (Online Analytical Processing) handles complex analytical queries on historical data — business intelligence, reporting.

Q140 What is a database schema?

Answer: A schema is the logical blueprint of a database: tables, columns, data types, constraints, relationships, and indexes. It is defined before populating the database and can be visualized via ERD. Changing schema in production requires careful migration.

Q141 What are SQL aggregate functions?

Answer: Aggregate functions compute values from multiple rows: COUNT() (number of rows), SUM() (total), AVG() (average), MAX() (highest), MIN() (lowest). Used with GROUP BY to aggregate results by category.

Q142 What is GROUP BY in SQL?

Answer: GROUP BY groups rows with the same values in specified columns, allowing aggregate functions to be applied per group. Example: SELECT dept, COUNT(*) FROM employees GROUP BY dept — shows headcount per department.

Q143 What is a database trigger?

Answer: A trigger is a stored procedure that automatically executes in response to specific events (INSERT, UPDATE, DELETE) on a table. Used for audit logging, enforcing business rules, and maintaining data consistency automatically.

Q144 What is connection pooling?

Answer: Connection pooling maintains a cache of reusable database connections, eliminating the overhead of opening and closing connections for each query. Dramatically improves performance and scalability for web applications under load.

Q145 What is a self-join?

Answer: A self-join joins a table to itself using aliases, treating it as two separate tables. Useful for hierarchical data like employee-manager relationships in the same table. Example: finding all employees and their direct managers.

Q146 What is database replication?

Answer: Replication copies data from one database server to others in real-time or near-real-time. Improves read performance (read replicas), availability, and disaster recovery. Types: synchronous (guaranteed consistency), asynchronous (eventual consistency).

Q147 What is database sharding?

Answer: Sharding horizontally partitions data across multiple database servers (shards), each holding a subset of data. Enables massive scalability and improves write throughput. Used by large-scale applications like social media platforms.

Q148 What is an ORM?

Answer: ORM (Object-Relational Mapper) maps database tables to programming objects, allowing developers to query and manipulate data using their programming language instead of raw SQL. Examples: Hibernate (Java), SQLAlchemy (Python), Entity Framework (.NET).

Q149 What is caching in relation to databases?

Answer: Caching stores frequently accessed query results in fast memory (Redis, Memcached) to reduce database load. Cache hit: data found in cache; cache miss: data fetched from database and then stored in cache. Dramatically reduces response times.

Q150 What is big data?

Answer: Big data refers to extremely large datasets characterized by the 3 V's: Volume (massive size), Velocity (high-speed generation), and Variety (structured/unstructured/semi-structured). Processing frameworks: Apache Hadoop, Apache Spark. Storage: data lakes.

SECTION 06

Operating Systems

Q151 – Q180

Q151 What are the main functions of an operating system?

Answer: Process management (scheduling, multitasking), Memory management (RAM allocation/deallocation), File system management, Device management (drivers), Security and access control, User interface (CLI/GUI), and Network management.

Q152 What is the kernel of an OS?

Answer: The kernel is the core OS component managing system resources: CPU scheduling, memory, device drivers, and system calls. It runs in privileged mode (kernel space) with direct hardware access, isolated from user applications (user space).

Q153 What is the difference between GUI and CLI?

Answer: GUI (Graphical User Interface) uses windows, icons, and menus — intuitive for everyday users. Examples: Windows, macOS. CLI (Command Line Interface) uses text commands — more powerful, scriptable, and efficient for IT professionals. Examples: Linux Terminal, PowerShell.

Q154 What is Windows Task Manager?

Answer: Task Manager shows running processes, CPU/RAM/disk/network usage, startup impact programs, and active user sessions. Allows ending unresponsive processes and monitoring system performance. Launched with Ctrl+Shift+Esc or Ctrl+Alt+Delete.

Q155 What is the Windows Registry?

Answer: The Registry is a hierarchical database storing OS and application settings, hardware configuration, and user preferences. Key hives: HKEY_LOCAL_MACHINE, HKEY_CURRENT_USER. Edited via 'regedit'. Corruption can cause serious system instability.

Q156 What is a process in an OS?

Answer: A process is an instance of a running program with its own memory space, resources, and execution state. The OS tracks each process via a Process Control Block (PCB) storing process ID, state, priority, memory maps, and open files.

Q157 What is process scheduling?

Answer: Process scheduling determines which process receives CPU time. Algorithms: FCFS (First Come First Serve), Round Robin (equal time slices — prevents starvation), Priority Scheduling, and SJF (Shortest Job First). Goal: maximize CPU utilization and fairness.

Q158 What is deadlock?

Answer: Deadlock is a state where two or more processes are permanently blocked, each waiting for resources held by the other. Four necessary conditions (Coffman): Mutual Exclusion, Hold and Wait, No Preemption, and Circular Wait. All must exist simultaneously.

Q159 What is paging in memory management?

Answer: Paging divides physical memory into fixed-size frames and logical memory into equal-size pages. The OS uses a page table to map pages to frames. Eliminates external fragmentation and enables virtual memory to exceed physical RAM.

Q160 What is swapping?

Answer: Swapping moves idle or low-priority processes from RAM to disk (swap space/pagefile) to free RAM for active processes. Extends available memory but adds significant latency due to slow disk I/O compared to RAM access speeds.

Q161 What is Linux?

Answer: Linux is a free, open-source Unix-like OS. The kernel was created by Linus Torvalds in 1991. Available in distributions (distros): Ubuntu, CentOS, Red Hat, Debian, Kali Linux. Dominates servers, cloud, embedded systems, and supercomputers.

Q162 What is the difference between Linux and Windows?

Answer: Linux: open-source, free, highly customizable, more secure, command-line-first, dominant on servers. Windows: proprietary, user-friendly GUI, better desktop software compatibility, extensive gaming support. Linux uses ext4; Windows uses NTFS by default.

Q163 What are essential Linux commands?

Answer: ls (list files), cd (change directory), pwd (current directory), mkdir (make directory), rm (remove), cp (copy), mv (move/rename), cat (display file), grep (search text), chmod (permissions), sudo (run as root), ps (list processes), top (system monitor).

Q164 What is sudo in Linux?

Answer: sudo ('Superuser Do') allows permitted users to run specific commands with root (superuser) privileges without logging in as root. Safer than direct root login as it logs all elevated commands. Example: 'sudo apt update' updates package lists as root.

Q165 What is a shell in Linux?

Answer: A shell is a command-line interpreter between the user and the OS kernel, processing commands and scripts. Popular shells: Bash (most common default), Zsh, Fish, Dash. Shell scripting enables powerful automation of repetitive tasks.

Q166 What is file permission in Linux?

Answer: Linux permissions control read (r), write (w), and execute (x) access for owner, group, and others. Example: -rwxr-xr-- means owner has full access, group can read/execute, others can only read. Changed with chmod. Ownership managed with chown.

Q167 What is a cron job?

Answer: A cron job schedules automatic task execution at specified times via the cron daemon. Configured in the crontab file. Syntax: minute hour day month weekday command. Example: '0 2 * * * /backup.sh' runs a backup script every night at 2:00 AM.

Q168 What is Active Directory?

Answer: Active Directory (AD) is Microsoft's directory service managing users, computers, groups, and resources in Windows domains. Provides centralized authentication, authorization, Group Policy enforcement, and single sign-on (SSO). Essential in enterprise IT.

■ *Highly relevant to your Senior IT Engineer role at Chandan Steel!*

Q169 What is Group Policy in Windows?

Answer: Group Policy is an AD feature defining and enforcing configurations for users and computers in a domain: security policies, software deployment, desktop restrictions, mapped drives, and startup scripts. Managed via Group Policy Management Console (GPMC).

Q170 What is UAC (User Account Control)?

Answer: UAC prevents unauthorized changes to Windows by prompting for confirmation or credentials when actions require admin privileges. It runs most applications with standard user rights even for admin accounts, reducing malware damage potential.

Q171 What is Safe Mode in Windows?

Answer: Safe Mode starts Windows with minimal drivers and services — no startup programs, no third-party drivers. Useful for removing malware, fixing driver conflicts, restoring corrupted system files, and troubleshooting issues that prevent normal boot.

Q172 What is Disk Management in Windows?

Answer: Disk Management is a built-in utility for managing hard drives: creating, deleting, and formatting partitions; changing drive letters; extending or shrinking volumes; and initializing new disks. Accessed via Computer Management console.

Q173 What is Windows Event Viewer?

Answer: Event Viewer logs system events: Application (app errors/info), Security (login attempts, policy changes), System (OS and hardware events). Invaluable for troubleshooting — filter by Error/Critical level and correlate timestamps with issues.

Q174 What is Task Scheduler?

Answer: Task Scheduler automates programs or scripts at specific times or triggered by system events (startup, login, idle, event log entry). Used for automated backups, maintenance scripts, and monitoring without user intervention.

Q175 What is the Windows boot sequence?

Answer: Power on □ POST (BIOS/UEFI hardware check) □ Boot device selection □ Boot loader loads (Windows Boot Manager/GRUB) □ OS kernel loads into memory □ System services and drivers initialize □ Login screen presented to user.

Q176 What is a system restore point?

Answer: A restore point is a Windows snapshot of system files, registry settings, and installed programs at a specific time. If problems arise after updates or installations, Windows can roll back to a previous state. Does not affect personal user files.

Q177 What is memory fragmentation?

Answer: Fragmentation scatters free memory in non-contiguous blocks. Internal fragmentation wastes space within allocated blocks. External fragmentation has sufficient total free memory but lacks a single contiguous block large enough. Paging and compaction address this.

Q178 What is an interrupt in OS?

Answer: An interrupt is a hardware or software signal requesting immediate CPU attention, suspending the current task. Hardware interrupts: from devices (keyboard press, network packet). Software interrupts: from programs (system calls). Handled by an ISR (Interrupt Service Routine).

Q179 What is the difference between 32-bit and 64-bit OS?

Answer: 32-bit OS supports maximum 4 GB RAM (2^{32} addresses). 64-bit OS supports terabytes of RAM (2^{64} addresses), runs 64-bit optimized code faster, and provides better security features. All modern OSes and applications are 64-bit.

Q180 What is a Windows service?

Answer: A Windows service is a long-running background process that starts automatically or on demand, often with no user interface. Examples: Windows Update, Print Spooler, DHCP Client, DNS Client. Managed via Services.msc console.

SECTION 07

Hardware & Devices

Q181 – Q210

Q181 What are the main components inside a desktop PC?

Answer: CPU (processor), Motherboard (main circuit board), RAM (temporary memory), Storage (HDD/SSD), PSU (power supply), GPU (graphics), CPU cooler (heatsink/fan), NIC (network card), case/chassis, and optional: optical drive, sound card.

Q182 What is a power supply unit (PSU)?

Answer: PSU converts AC mains power to regulated DC voltages (+12V, +5V, +3.3V) for computer components. Wattage determines capacity. Efficiency ratings: 80 Plus Bronze/Silver/Gold/Platinum. Modular PSUs allow removing unused cables for better airflow.

Q183 What is RAID and its types?

Answer: RAID (Redundant Array of Independent Disks) combines drives for performance or fault tolerance. RAID 0: striping (speed, no redundancy). RAID 1: mirroring (full redundancy). RAID 5: striping + distributed parity (balance). RAID 10: mirroring + striping (best of both).

Q184 What is the difference between laser and inkjet printer?

Answer: Laser printers use toner (powder) fused by heat — fast, high volume, lower cost per page for text. Inkjet printers use liquid ink — better photo quality, lower upfront cost but higher cost per page. Laser better for office; inkjet better for photos.

Q185 What is USB and its versions?

Answer: USB (Universal Serial Bus) is the standard peripheral interface. Versions: USB 2.0 (480 Mbps), USB 3.0 (5 Gbps, blue port), USB 3.1 (10 Gbps), USB 3.2 (20 Gbps), USB4 (40 Gbps). Connectors: Type-A, Type-B, micro/mini, Type-C (reversible).

Q186 What is HDMI?

Answer: HDMI (High-Definition Multimedia Interface) transmits digital video and audio through a single cable. Used to connect computers, gaming consoles, and players to displays. HDMI 2.1 supports 8K/60Hz and 4K/120Hz. Standard in consumer electronics.

Q187 What is a UPS?

Answer: UPS (Uninterruptible Power Supply) provides battery backup power when mains power fails or fluctuates. Protects against surges, brownouts, and outages. Critical for servers and workstations to prevent data loss and allow graceful shutdown.

Q188 What is thermal paste?

Answer: Thermal paste (thermal compound) fills microscopic gaps between the CPU surface and heatsink, improving heat transfer conductivity. Significantly reduces CPU temperatures. Should be replaced when removing/reinstalling a CPU cooler.

Q189 What is overclocking?

Answer: Overclocking increases a CPU or GPU's clock speed beyond manufacturer-rated specifications for additional performance. Generates more heat and requires better cooling. May void warranty and reduce lifespan if voltage is increased unsafely.

Q190 What is the CMOS battery?

Answer: The CMOS battery (typically CR2032 coin cell) on the motherboard powers the CMOS chip maintaining BIOS settings and real-time clock (RTC) when the PC is powered off. When it fails, the system loses date/time and reverts to default BIOS settings.

Q191 What is an M.2 SSD?

Answer: M.2 is a compact SSD form factor plugging directly into the motherboard M.2 slot. It supports both SATA (up to 550 MB/s) and NVMe/PCIe interfaces (up to 7000+ MB/s). NVMe M.2 SSDs offer dramatically faster storage than traditional SATA SSDs.

Q192 What is the difference between IDE and SATA?

Answer: IDE (Parallel ATA) is an older interface with wide ribbon cables, supporting up to 133 MB/s. SATA (Serial ATA) uses thin cables, supports faster speeds (SATA III = 600 MB/s), hot-swapping, and Native Command Queuing. All modern storage uses SATA or NVMe.

Q193 What is DisplayPort?

Answer: DisplayPort is a digital display interface primarily connecting computers to monitors. Supports higher refresh rates and resolutions than older HDMI versions. Features multi-monitor daisy-chaining. DisplayPort 2.1 supports 8K at 60Hz. Common in professional and gaming monitors.

Q194 What are LCD panel types?

Answer: TN (Twisted Nematic): fastest response, lowest cost, narrow viewing angles and poor colors. IPS (In-Plane Switching): excellent color accuracy and wide viewing angles, ideal for design. VA (Vertical Alignment): best contrast ratio, between TN and IPS for most uses.

Q195 What is Bluetooth?

Answer: Bluetooth is short-range wireless technology for device communication over ~10-100 meters using 2.4 GHz band. Used for headphones, keyboards, mice, speakers, smartwatches. Bluetooth 5.0+ offers 2x speed, 4x range, and 8x broadcast capacity.

Q196 What is NFC?

Answer: NFC (Near Field Communication) enables wireless data exchange at very short distances (up to 4 cm), requiring devices to nearly touch. Used for contactless payments (Google Pay, Apple Pay), access cards, transit passes, and rapid device pairing.

Q197 What is a KVM switch?

Answer: KVM (Keyboard, Video, Mouse) switch lets one set of keyboard, monitor, and mouse control multiple computers. Switch between systems via button press or keyboard shortcut. Used in server rooms and by power users managing multiple PCs.

Q198 What is a CPU socket?

Answer: A CPU socket is the physical connector on the motherboard where the processor is installed. Socket types are manufacturer-specific and generation-specific: Intel uses LGA (Land Grid Array); AMD uses PGA AM4 or LGA AM5. CPU and socket must be compatible.

Q199 What are CPU cores and hyper-threading?

Answer: A CPU core is an independent processing unit. Multi-core CPUs (quad, octa) execute multiple tasks simultaneously. Hyper-threading (Intel) / SMT (AMD) lets each core handle 2 threads, effectively doubling logical processors for the OS.

Q200 What is e-waste and responsible disposal?

Answer: E-waste is discarded electronics containing toxic materials (lead, mercury, cadmium). Must be disposed through certified e-waste recyclers — not regular trash. IT professionals should follow organizational ITAD (IT Asset Disposition) policies and WEEE regulations.

Q201 What is a touchscreen and its types?

Answer: A touchscreen detects touch input. Types: Resistive (pressure-based, cheap, stylus-friendly), Capacitive (uses electrical charge, multi-touch, most smartphones), Infrared, and Surface Acoustic Wave. Capacitive dominates modern smartphones and tablets.

Q202 What is a barcode scanner?

Answer: Barcode scanners read barcodes using laser or image sensors. 1D scanners read traditional barcodes. 2D scanners read QR codes and DataMatrix. Omnidirectional scanners read from any angle (retail checkouts). Widely used in retail, warehouses, and healthcare.

Q20
3

What is a solid-state hybrid drive (SSHD)?

Answer: SSHD combines HDD mechanical storage with a small SSD cache (4-8 GB). Frequently accessed data is automatically cached in the fast SSD portion. Offers better performance than pure HDD at a lower price than pure SSD for bulk storage needs.

Q20
4

What is a sound card?

Answer: A sound card processes audio input and output for a computer. Onboard audio is integrated into modern motherboards. Dedicated sound cards offer superior audio quality for recording, music production, and audiophile listening. Connect via 3.5mm or digital optical.

Q20
5

What is network cable and its categories?

Answer: Network cables: Cat5e (1 Gbps, 100m), Cat6 (1-10 Gbps, 55m), Cat6a (10 Gbps, 100m), Cat7 (10+ Gbps, shielded). Fiber optic (single-mode/multi-mode) for high-speed long-distance. Cat6 is current standard for enterprise LAN installations.

Q20
6

What is VGA, DVI, and HDMI?

Answer: VGA is an analog video connector (max 2560x1600) — older technology, no audio. DVI carries digital/analog video (max 2560x1600) — no audio. HDMI carries digital video and audio (4K/8K support). VGA and DVI are being phased out in favor of HDMI/DisplayPort.

Q207

What is a projector and its types?

Answer: Projectors display images/video on large surfaces. Technologies: LCD (bright, good color), DLP (sharp images, compact), LCoS (highest resolution, expensive), Laser (long life, instant on). Key specs: lumens (brightness), resolution, throw ratio, contrast ratio.

Q20
8

What is a GPU and when is it needed?

Answer: GPU (Graphics Processing Unit) processes graphics and parallel computations. Dedicated GPUs are essential for gaming, 3D rendering, video editing, CAD design, machine learning, and scientific computing. Integrated GPUs suffice for everyday office tasks.

Q20
9

What is wireless charging?

Answer: Wireless charging uses electromagnetic induction to transfer energy between a charging pad and compatible device. Based on Qi standard. Slower than wired charging but convenient. Found in smartphones, smartwatches, and earbuds. Power up to 15W (standard) or 65W+ (fast).

Q210

What is a smartwatch?

Answer: A smartwatch is a wearable computing device extending smartphone functionality to the wrist: notifications, health monitoring (heart rate, SpO2, ECG), GPS, payments, fitness tracking. Examples: Apple Watch, Samsung Galaxy Watch, Garmin.

SECTION 08

Troubleshooting & Support

Q211 – Q240

Q211 What is the CompTIA troubleshooting methodology?

Answer: 1) Identify the problem (gather information, symptoms), 2) Establish theory of probable cause, 3) Test the theory, 4) Establish an action plan, 5) Implement the solution, 6) Verify full system functionality, 7) Document findings and solution.

■ *This exact 7-step methodology appears directly on the CompTIA ITF+ exam.*

Q212 What does it mean when a PC won't power on?

Answer: Checklist: power cable seated, outlet working (test with another device), PSU power switch on, PSU wattage adequate, 24-pin motherboard connector fully seated, 4/8-pin CPU power connected, RAM properly installed, check for BIOS POST beep codes.

Q213 What is the Blue Screen of Death (BSOD)?

Answer: BSOD is a Windows critical stop error when the OS encounters a fatal fault it cannot recover from. Shows a stop code (e.g., IRQL_NOT_LESS_OR_EQUAL, MEMORY_MANAGEMENT). Diagnose using Event Viewer and WinDbg to analyze minidump files.

Q214 How do you troubleshoot slow computer performance?

Answer: Check Task Manager for high CPU/RAM/disk usage. Scan for malware. Check startup programs. Run chkdsk (disk errors). Check available disk space (free at least 15%). Monitor temperatures (overheating throttles performance). Consider RAM or SSD upgrade.

Q215 What is POST and what do beep codes indicate?

Answer: POST (Power-On Self Test) is the hardware diagnostic run at startup by BIOS/UEFI. If POST fails, the system emits beep codes. Common: 1 short = normal, 3 short = RAM error, continuous = power issue, no beep with no display = failed CPU or motherboard.

Q216 How do you troubleshoot no internet connectivity?

Answer: Step-by-step: 1) Check physical cable/Wi-Fi connection, 2) ipconfig — is IP 169.254.x.x? (DHCP failure), 3) ping 127.0.0.1 (loopback test), 4) ping default gateway, 5) ping 8.8.8.8 (external IP), 6) ping google.com (DNS test), 7) check firewall rules, 8) contact ISP.

Q217 What is the ipconfig command?

Answer: ipconfig shows network configuration: IP address, subnet mask, default gateway, DNS server. Key switches: /all (detailed including MAC, DHCP server), /release (release DHCP lease), /renew (request new DHCP lease), /flushdns (clear DNS resolver cache).

Q218 What is nslookup?

Answer: nslookup queries DNS servers for domain name resolution. Use it to verify DNS is working: 'nslookup google.com' should return Google's IP addresses. Change DNS server in nslookup to test alternate DNS and diagnose DNS resolution failures.

Q219 What is tracert / traceroute?

Answer: tracert (Windows) / traceroute (Linux) traces the network path packets take to a destination, showing each router hop with round-trip latency. Identifies where delays or failures occur. High latency at a specific hop indicates a congestion or routing issue.

Q220 How do you troubleshoot a printer?

Answer: Check: power and cables (USB/network), ink/toner levels, paper jam, print a test page from the printer panel, check print queue for stuck jobs (clear it), reinstall/update printer driver, verify correct default printer selected, check Windows Print Spooler service is running.

Q221 What causes PC overheating?

Answer: Causes: failed or blocked cooling fans, dust buildup on heatsinks and vents, expired/dried-out thermal paste, inadequate case airflow, overclocking without sufficient cooling. Prevention: regular compressed air cleaning, proper case fan placement, quality thermal paste.

Q222 What is chkdsk?

Answer: chkdsk (Check Disk) scans the file system and disk surface for errors, cross-linked files, and bad sectors. 'chkdsk C: /f' fixes file system errors; '/r' also locates bad sectors and recovers readable data. Run from elevated Command Prompt; may require reboot for system drive.

Q223 What is SFC (System File Checker)?

Answer: SFC (sfc /scannow) scans Windows system files for corruption and replaces damaged files from a cached copy. Run as Administrator. If SFC reports errors it cannot fix, run DISM first to repair the component store, then run SFC again.

Q224 What is DISM?

Answer: DISM (Deployment Image Servicing and Management) repairs the Windows system image and component store. Key command: 'DISM /Online /Cleanup-Image /RestoreHealth'. Run this before SFC when the system is seriously corrupted, as it downloads clean files from Windows Update.

Q225 How do you remove malware?

Answer: Disconnect from internet. Boot into Safe Mode. Run full scan with updated antivirus. Use Malwarebytes as secondary scanner. Check and remove suspicious startup entries (msconfig/Task Scheduler/Registry). Check browser extensions. Restore from clean backup if necessary.

Q226 What does 'No bootable device found' error mean?

Answer: The system cannot locate a bootable OS. Causes: wrong boot order in BIOS/UEFI (check storage device listed first), corrupted boot record (MBR/GPT), failed storage device, or missing OS. Fix: check boot order, run Windows Startup Repair, or rebuild BCD with 'bootrec /rebuildbcd'.

Q227 What is Device Manager?

Answer: Device Manager displays all installed hardware and driver status. Yellow exclamation = driver problem. Red X = disabled device. Used to update, roll back, or uninstall drivers; enable/disable devices; view hardware IDs; and troubleshoot resource conflicts.

Q228 What is a help desk ticket / incident ticket?

Answer: A help desk ticket formally records a user's IT issue: description, priority (P1-P4), category, assigned technician, SLA timer, status, resolution steps, and closure notes. Enables tracking, accountability, reporting, and knowledge base building. Tools: ServiceNow, Jira.

Q229 What is Remote Desktop Protocol (RDP)?

Answer: RDP (port 3389) allows connecting to and controlling a Windows computer over a network as if sitting at it. Used for remote IT support, remote work, and server administration. Best practice: disable when not needed; use VPN + NLA authentication.

Q230 What is Windows Memory Diagnostic?

Answer: Windows Memory Diagnostic (mdsched.exe) tests RAM for errors by running memory pattern tests during reboot. Useful when experiencing random BSODs, application crashes, or data corruption that may indicate faulty RAM modules. Results appear in Event Viewer after reboot.

Q231 What is a clean boot?

Answer: A clean boot starts Windows with only essential Microsoft services and no third-party startup programs. Configured via msconfig. Used to diagnose whether a third-party application or service causes system instability, crashes, or performance issues.

Q232 How do you fix a 'disk full' error?

Answer: Run Disk Cleanup utility, delete temp files (%temp%, C:\Windows\Temp), empty Recycle Bin, uninstall unused programs, disable/delete Windows hibernation file (hiberfil.sys), manage Shadow Copy (VSS) storage, move files to external drive, or upgrade storage.

Q233 What is Event Log used for?

Answer: Event Logs record system, application, and security events chronologically. Filter by Error/Critical level and correlate with the time a problem occurred. Key IDs: Event 41 (unexpected shutdown), 1001 (system crash), 7031 (service crashed). Essential for root cause analysis.

Q234 What is a Windows recovery drive?

Answer: A recovery drive is a bootable USB containing Windows recovery tools: Startup Repair, System Restore, Reset PC, and Command Prompt access. Created via 'Create a recovery drive' in Windows settings. Essential for repairing systems that fail to boot.

Q235 What is thermal throttling?

Answer: Thermal throttling is automatic CPU/GPU clock speed reduction to prevent damage when temperatures exceed safe thresholds. Causes sudden performance drops during sustained workloads. Solution: clean heatsinks, replace thermal paste, improve case airflow.

Q236 What is a driver conflict?

Answer: A driver conflict occurs when drivers compete for the same system resources (IRQ, memory addresses), causing instability or device malfunction. Identified in Device Manager. Resolved by updating, rolling back, or reinstalling conflicting drivers.

Q237 How do you troubleshoot a USB device not recognized?

Answer: Try different USB port. Try device on another computer. Update USB host controller driver. Uninstall and reinstall device in Device Manager. Check Power Management settings (disable 'Allow computer to turn off this device to save power'). Check for USB selective suspend issues.

Q23
8**What is the importance of documentation in IT?**

Answer: Documentation records solutions, configurations, network diagrams, change history, and procedures. Benefits: faster problem resolution, knowledge transfer to new staff, audit compliance, consistent support quality, and reduced single-person dependency (bus factor).

Q23
9**What is a POST card / diagnostic card?**

Answer: A POST diagnostic card is a hardware tool that plugs into the motherboard to display POST codes as hexadecimal numbers, identifying where the boot process fails even when there is no video output. Used for deep hardware-level troubleshooting.

Q24
0**What is the difference between reboot, shutdown, and restart?**

Answer: Shutdown powers off the PC completely (with Fast Startup in modern Windows, it may hibernate the kernel). Restart performs a full cycle: complete shutdown then power-on — loads a fresh OS state. In troubleshooting, always use Restart (not Shutdown) to fully clear system state.

SECTION 09

Digital Literacy & Devices

Q241 – Q270

Q241 What is digital literacy?

Answer: Digital literacy is the ability to effectively use, evaluate, and create information using digital technologies. It includes operating computers, using the internet, understanding digital security, evaluating online information credibility, and protecting privacy.

Q242 What is the difference between Android and iOS?

Answer: Android (Google) is open-source, runs on many manufacturers' devices, highly customizable, large app variety. iOS (Apple) is closed-source, exclusive to iPhone/iPad, known for strong security, privacy controls, regular long-term updates, and seamless ecosystem integration.

Q243 What is cloud storage?

Answer: Cloud storage stores data on remote servers accessible via the internet from any device. Benefits: automatic backup, anywhere access, easy sharing, and no local storage capacity concerns. Examples: Google Drive, OneDrive, Dropbox, iCloud. Security depends on the provider.

Q244 What is a QR code?

Answer: QR (Quick Response) code is a 2D matrix barcode storing up to 7,000 characters of data, readable by smartphone cameras. Used for URLs, payment links, product info, menus, and event tickets. Much higher data capacity than 1D barcodes.

Q245 What is GPS?

Answer: GPS (Global Positioning System) uses a network of 24+ satellites to determine geographic location with 3-5 meter accuracy for consumer devices. Used in navigation, fleet tracking, fitness apps, augmented reality, and location-based services.

Q24
6

What is VoIP?

Answer: VoIP (Voice over Internet Protocol) transmits voice over the internet instead of traditional telephone networks. Benefits: lower cost (especially international calls), video calling integration, portability. Examples: WhatsApp, Skype, Zoom, Microsoft Teams, Cisco Webex.

Q247

What is e-commerce?

Answer: E-commerce is buying and selling via the internet. Types: B2C (Amazon, Flipkart), B2B (Alibaba), C2C (eBay, OLX), D2C (brand sells directly). Requires secure payment gateways, SSL/HTTPS, and compliance with consumer data protection laws.

Q248

What is a digital signature?

Answer: A digital signature uses asymmetric cryptography to verify document authenticity and integrity. The sender signs with their private key; recipients verify with the sender's public key. Legally binding in India (IT Act 2000) and most countries. Cannot be repudiated.

Q24
9

What is a web cookie?

Answer: A web cookie is a small text file stored by a browser containing session info, preferences, or login state. First-party cookies improve user experience (remember login). Third-party cookies track users across sites for advertising. GDPR requires cookie consent.

Q25
0

What is browser cache?

Answer: Browser cache stores web assets (images, HTML, CSS, JS) locally for faster page loading on repeat visits. Reduces server load and bandwidth. Stale cached data can cause display issues — clearing the cache forces fresh downloads of all page assets.

Q251 What is social media literacy?

Answer: Social media literacy is understanding how to use platforms responsibly: recognizing misinformation, protecting privacy, understanding algorithms, digital footprint awareness, appropriate sharing, and recognizing the psychological effects of social media usage.

Q252 What is cyberbullying?

Answer: Cyberbullying uses digital technology to harass, threaten, or intimidate others via social media, messaging apps, gaming, or email. It causes serious psychological harm. Report to the platform, school/employer, and law enforcement (IT Act Section 66A in India).

Q253 What is net neutrality?

Answer: Net neutrality is the principle that ISPs must treat all internet traffic equally — no throttling specific services, no paid priority lanes, no blocking competitors' content. It preserves a level playing field for all content creators and services online.

Q254 What is digital privacy?

Answer: Digital privacy is the right to control personal information shared online. Best practices: use strong unique passwords, enable MFA, review app permissions, use encrypted messaging (Signal), avoid public Wi-Fi without VPN, read privacy policies before sharing data.

Q255 What is the dark web vs deep web?

Answer: Deep web: all internet content not indexed by search engines (email inboxes, private databases, banking portals) — most internet by volume. Dark web: a small subset of deep web requiring Tor browser, intentionally hidden. Contains both legitimate and illegal activity.

Q256 What is IoT (Internet of Things)?

Answer: IoT connects physical devices embedded with sensors, software, and connectivity to collect and exchange data. Examples: smart thermostats, IP cameras, wearables, connected appliances. IoT security is critical as these devices often lack robust protection.

Q257 What is AI (Artificial Intelligence)?

Answer: AI is computer systems performing tasks requiring human-like intelligence: reasoning, learning, pattern recognition, language understanding. Types: Narrow AI (specific tasks — image recognition, chatbots), General AI (theoretical), Superintelligence (hypothetical). Powered by ML.

Q258 What is machine learning?

Answer: Machine learning is AI where algorithms learn from data to make predictions or decisions without explicit programming for each task. Types: Supervised (labeled data), Unsupervised (pattern discovery), Reinforcement (reward-based learning). Powers recommendations, fraud detection, and spam filtering.

Q259 What is augmented reality (AR)?

Answer: AR overlays digital information onto the real world using devices like smartphones or smart glasses. Examples: Google Maps Live View navigation, IKEA furniture placement app, Snapchat filters, industrial maintenance guides. Different from VR (fully virtual environments).

Q260 What is virtual reality (VR)?

Answer: VR immerses users in a fully digital environment using headsets (Meta Quest, PlayStation VR). Applications: gaming, medical/military training simulations, virtual tourism, therapy (phobia treatment), and architectural visualization. Requires high-performance GPU for smooth 90+ fps.

Q261 What is 5G technology?

Answer: 5G is the fifth generation of mobile networks offering speeds up to 10 Gbps, sub-1ms latency, and massive device density. Enables autonomous vehicles, smart cities, industrial IoT, cloud gaming, and remote surgery. Uses mmWave, sub-6GHz, and network slicing.

Q26
2

What is blockchain?

Answer: Blockchain is a distributed, immutable ledger recording transactions in linked blocks secured by cryptographic hashing. It is decentralized (no single authority), transparent, and tamper-resistant. Used in cryptocurrency (Bitcoin), supply chain, healthcare records, and smart contracts.

Q26
3

What is a chatbot?

Answer: A chatbot simulates human conversation via text or voice using NLP (Natural Language Processing). Rule-based bots follow scripts; AI-powered bots (like ChatGPT) generate contextual responses. Used in customer service, virtual assistants, and IT helpdesks.

Q26
4

What is open data?

Answer: Open data is freely available data anyone can use, modify, and share without restriction. Governments and organizations publish open datasets for transparency and innovation. Examples: data.gov, data.gov.in (India), OpenStreetMap, WHO health datasets.

Q26
5

What is data mining?

Answer: Data mining discovers hidden patterns, correlations, and anomalies in large datasets using statistical and ML techniques. Used in marketing (customer segmentation), fraud detection, medical diagnosis, and business intelligence to extract actionable insights.

Q26
6

What is digital footprint?

Answer: A digital footprint is the trail of data left by online activity: browsing history, social media posts, purchases, location data, and app usage. Permanent footprints are intentional (posted content); passive footprints are collected without direct action (cookies, logs).

Q267 What is screen time management?

Answer: Screen time management involves monitoring and limiting device usage to maintain work-life balance and reduce eye strain, sleep disruption, and sedentary behavior. Features: iOS Screen Time, Android Digital Wellbeing, parental controls, blue light filters, and 20-20-20 rule.

Q268 What is digital accessibility?

Answer: Digital accessibility ensures technology is usable by people with disabilities. Includes screen readers (for blind users), closed captions (deaf users), keyboard-only navigation, high contrast modes, and voice control. Mandated by WCAG guidelines and disability laws.

Q269 What is the Internet of Everything (IoE)?

Answer: IoE expands IoT by connecting people, processes, data, and devices into intelligent networks. Moving beyond sensor-to-server, IoE creates human-to-machine and human-to-human connections for smarter cities, healthcare, and industry. Coined by Cisco.

Q270 What is a smart home?

Answer: A smart home uses internet-connected IoT devices for automated control of lighting, HVAC, security, entertainment, and appliances — controlled via smartphone apps or voice assistants (Alexa, Google Home). Energy efficiency and security are key benefits and concerns.

SECTION 10

Cloud, Virtualization & Career

Q271 – Q300

Q271 What is hybrid cloud?

Answer: Hybrid cloud integrates private cloud (on-premises infrastructure) with public cloud services, connected via encrypted links. Sensitive/regulated data stays in private cloud; scalable workloads use public cloud. Offers flexibility, cost optimization, and compliance capabilities.

Q272 What is multi-cloud?

Answer: Multi-cloud uses services from multiple public cloud providers (AWS + Azure + GCP) to avoid vendor lock-in, leverage best-of-breed services, improve resilience, and optimize costs. Challenge: increased management complexity and need for cloud-agnostic tools.

Q273 What is containerization?

Answer: Containerization packages applications and their dependencies into portable containers sharing the host OS kernel. Containers are lightweight, start fast, and run identically across environments. Docker is the dominant container platform; Kubernetes orchestrates containers at scale.

Q274 What is Docker?

Answer: Docker is the leading container platform for building, shipping, and running applications. Core concepts: Image (immutable template), Container (running instance), Dockerfile (build instructions), Docker Compose (multi-container apps), Docker Hub (public image registry).

Q275 What is Kubernetes?

Answer: Kubernetes (K8s) is an open-source container orchestration system automating deployment, scaling, load balancing, and self-healing of containerized applications. It manages clusters of container nodes. Created by Google and donated to CNCF in 2014.

Q276 What is serverless computing?

Answer: Serverless lets developers run code without provisioning or managing servers. The cloud provider handles all infrastructure, auto-scaling, and billing per execution. Examples: AWS Lambda, Azure Functions, Google Cloud Functions. Ideal for event-driven, sporadic workloads.

Q277 What is edge computing?

Answer: Edge computing processes data near its source (IoT devices, factories, retail stores) rather than sending everything to centralized cloud data centers. Reduces latency, conserves bandwidth, and improves real-time responsiveness for autonomous vehicles, drones, and industrial automation.

Q278 What is a data center?

Answer: A data center houses computing infrastructure: servers, storage arrays, networking equipment, power systems (UPS, generators), and precision cooling. Tiers: Tier 1 (99.671% uptime) to Tier 4 (99.995% uptime, fully fault-tolerant). PUE (Power Usage Effectiveness) measures efficiency.

Q279 What is disaster recovery (DR)?

Answer: DR is policies and procedures to restore IT systems after disasters. Key metrics: RTO (Recovery Time Objective — maximum tolerable downtime) and RPO (Recovery Point Objective — maximum acceptable data loss). DR strategies range from cold standby to hot standby active-active.

Q280 What is business continuity planning (BCP)?

Answer: BCP ensures critical business functions continue during/after disasters. IT component includes DR, backup systems, failover infrastructure, and incident response plans. BIA (Business Impact Analysis) identifies critical systems and acceptable downtime for each.

Q281 What is a CDN?

Answer: CDN (Content Delivery Network) distributes web content from servers geographically close to users, reducing latency and improving load times. Handles traffic spikes, DDoS mitigation, and SSL termination. Examples: Cloudflare, AWS CloudFront, Akamai, Azure CDN.

Q282 What is SD-WAN?

Answer: SD-WAN (Software-Defined WAN) uses software to manage and optimize WAN connections across multiple links (MPLS, broadband, LTE). Provides centralized control, dynamic path selection, and cost reduction. Popular for connecting branch offices to cloud applications.

Q283 What is DevSecOps?

Answer: DevSecOps integrates security into every stage of the DevOps pipeline — 'shift left' security. Includes automated SAST/DAST scanning, container image scanning, dependency checking, secrets management, and compliance-as-code in CI/CD workflows.

Q284 What is Infrastructure as Code (IaC)?

Answer: IaC manages and provisions infrastructure through version-controlled code files instead of manual processes. Benefits: consistency, repeatability, peer review, and automation. Tools: Terraform (multi-cloud), AWS CloudFormation, Ansible, Puppet, Chef, Pulumi.

Q285 What is microservices architecture?

Answer: Microservices decomposes applications into small, independently deployable services communicating via APIs. Each service owns its data store and can be scaled independently. Contrasts with monolithic architecture. Enables faster releases but increases operational complexity.

Q286 What is CI/CD?

Answer: CI (Continuous Integration) automatically builds and tests code when developers push changes. CD (Continuous Delivery/Deployment) automatically deploys tested code to staging or production. Together they reduce release risk, increase velocity, and improve code quality. Tools: Jenkins, GitHub Actions, GitLab CI.

Q287 What certifications are recommended after CompTIA ITF+?

Answer: Career path: CompTIA A+ (hardware/software) □ CompTIA Network+ (networking) □ CompTIA Security+ (cybersecurity, DoD-approved). Specializations: Cloud+ (cloud), CySA+ (analyst), Pentest+ (pen testing). Also consider: Microsoft AZ-900, AWS Cloud Practitioner, Cisco CCNA.

■ *Direct guidance for your IT career progression!*

Q288 What is the CompTIA A+ certification?

Answer: CompTIA A+ (Core 1: 1101 + Core 2: 1102) covers hardware, OS, networking, security, and troubleshooting for desktops, laptops, and mobile devices. It is the most widely recognized entry-level IT certification globally. Highly valued by employers for IT support roles.

Q289 What is the ITIL framework?

Answer: ITIL (IT Infrastructure Library) v4 provides best practices for IT service management (ITSM). Key concepts: Service Value System, value streams, and four dimensions. ITIL Foundation certification is widely required for IT service management, help desk, and ITSM roles.

Q290 What is SLA (Service Level Agreement)?

Answer: An SLA is a contract defining expected service levels between provider and customer: uptime guarantees, response/resolution times, and penalties. Example: 99.9% uptime = max 8.76 hours downtime/year. SLAs drive IT team priorities and customer expectations.

Q291 What is change management in IT?

Answer: IT change management systematically manages modifications to IT systems to minimize risk and disruption. Process: change request, risk assessment, CAB (Change Advisory Board) approval, implementation, post-implementation review, and closure. Core ITIL process.

Q29
2

What is an IT ticketing system?

Answer: Ticketing systems manage support requests, incidents, and service requests: creation, assignment, prioritization, SLA tracking, escalation, knowledge base integration, and reporting. Examples: ServiceNow (enterprise), Jira Service Management, Zendesk, Freshdesk.

Q29
3

What is GDPR?

Answer: GDPR (General Data Protection Regulation) is the EU law governing personal data protection. Key requirements: lawful basis for processing, data subject rights (access, erasure, portability), 72-hour breach notification, privacy by design, and DPO appointment for high-risk processors.

Q29
4

What is an IT audit?

Answer: An IT audit evaluates IT infrastructure, policies, and controls for security, compliance, and operational efficiency. Auditors assess against standards: ISO 27001, SOC 2, HIPAA, PCI-DSS. Results identify risks, gaps, and recommendations for improvement.

Q29
5

What is the role of an IT administrator?

Answer: IT administrators manage organizational IT infrastructure: server maintenance, user account management (Active Directory), network monitoring, security patch management, backup management, hardware procurement, and Tier 2/3 technical support.

Q29
6

What skills are important for IT professionals in 2025?

Answer: High-demand IT skills: cloud computing (AWS/Azure), cybersecurity (Zero Trust, SIEM/SOAR), automation/scripting (Python, PowerShell), DevOps/CI-CD, AI/ML fundamentals, networking (SD-WAN, SASE), data analytics, communication skills, and continuous learning mindset.

■ *Relevant to your LinkedIn content on cybersecurity – great for posts!*

Q297 What is Green IT / Sustainable IT?

Answer: Green IT focuses on reducing the environmental impact of technology: energy-efficient hardware, server consolidation via virtualization, cloud migration (shared infrastructure), responsible e-waste disposal, paperless processes, and carbon-neutral data centers.

Q298 What is the future of IT?

Answer: Key trends: AI/ML everywhere, quantum computing, 6G networks, edge/fog computing, XR (extended reality), post-quantum cryptography, autonomous systems, sustainable computing, digital twins, and continued cloud-native transformation.

Q299 Why is continuous learning critical in IT?

Answer: IT evolves faster than almost any other field — technologies, threats, and platforms change constantly. Certifications, online learning (Coursera, Udemy, Pluralsight), reading documentation, building home labs, attending conferences, and contributing to communities all maintain relevance and career growth.

Q300 What advice would you give to someone starting an IT career?

Answer: Earn CompTIA ITF+ and A+ first. Build a home lab (old PCs, VirtualBox, Packet Tracer). Get comfortable with both Windows and Linux. Learn basic scripting (PowerShell, Bash). Build a strong LinkedIn presence. Contribute to open source. Network with IT professionals. Never stop learning — curiosity is your greatest IT asset.

■ *The best advice for any IT beginner — including this guide's reader!*